AI-Assisted DFT Automation using Quantum ESPRESSO & PWTK (2 to 6 Nov’26)

Dr. Aswathi Mohan T



LECTURE

Nudged Elastic Band Theory

Thermally activated process: Example

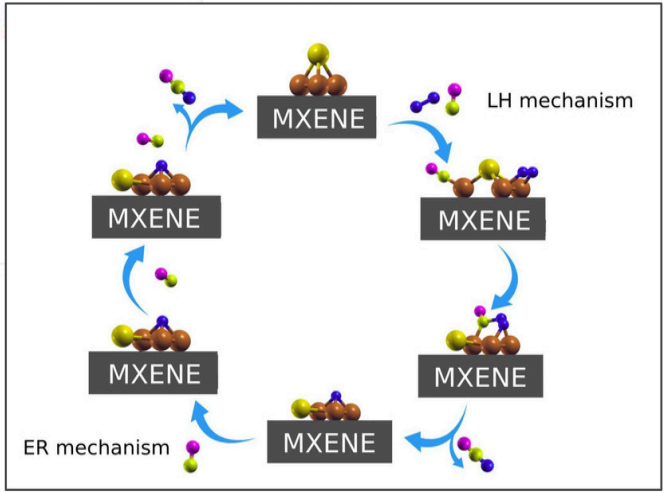
Issue 32, 2022



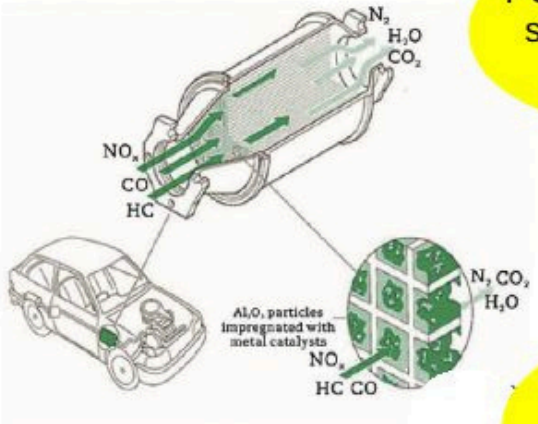
From the journal:
Physical Chemistry Chemical Physics

A low cost bimetallic $AuCu_3$ tetramer on Ti_2CO_2 MXene as an efficient catalyst for CO oxidation: a theoretical prediction†

Aswathi Mohan T. [a] and Prasenjit Ghosh *[bc]



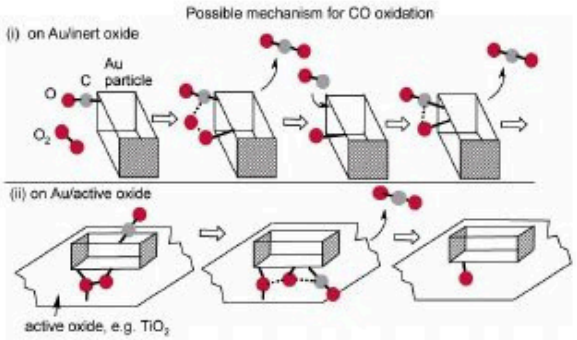
Motivation



PGM NPs on oxide suffers from cold start problem

Active support activates and/or supplies oxygen[1]

Au NPs on TiO_2 good for low T CO oxidation.





Au_7Cu_3 NCs on CeO_2(111), enhances the rate by 10^5 times.[2]

AuCu on SBA-15 had high stability and activity for CO oxidation. (TOF thrice that of Au[3])

1. J. Am. Chem. Soc. 124(49): 14770–14779, 2002.
2. J. Phys. Chem. Lett. 4: 2943–2947, 2013.
3. Chem. Commun., 27: 3187–3189, 2008.
4. J. Traffic Transp. Eng. (Engl. Ed.), 6 (5): 419-440, 2019.

Thermally activated process: Example

- The reaction proceeds in two steps as,
  $CO + 1/2\ O_2 \rightarrow CO_2$ (first cycle) followed by $CO + O \rightarrow CO_2$ (second cycle)

- Three mechanisms studied for CO oxidation;

1) LH – firstly CO and O_2 coadsorbs on the cluster and then reacts,



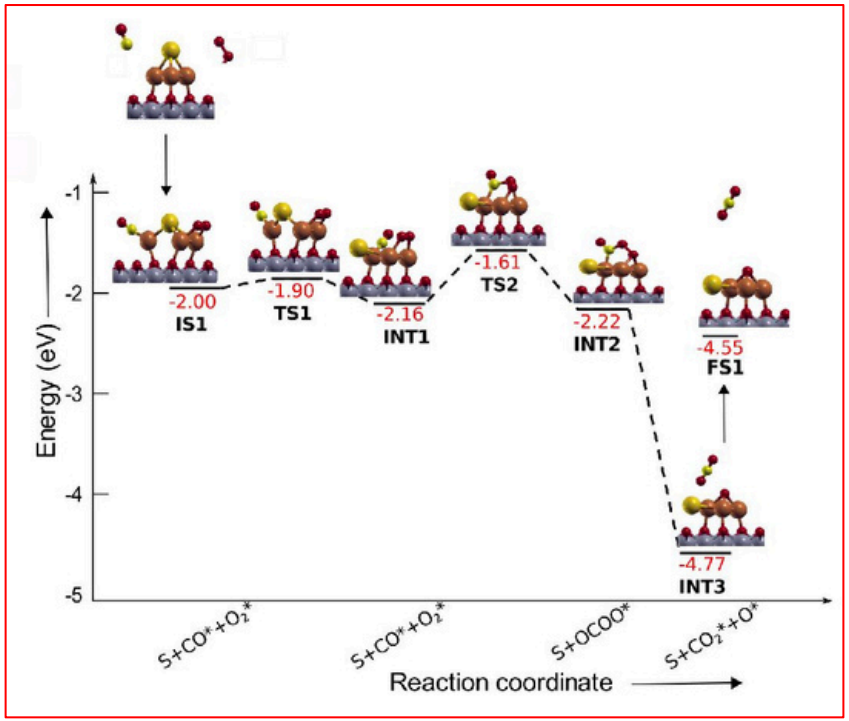
2) ER – firstly CO adsorbs on the cluster and then O_2 attacks it, and



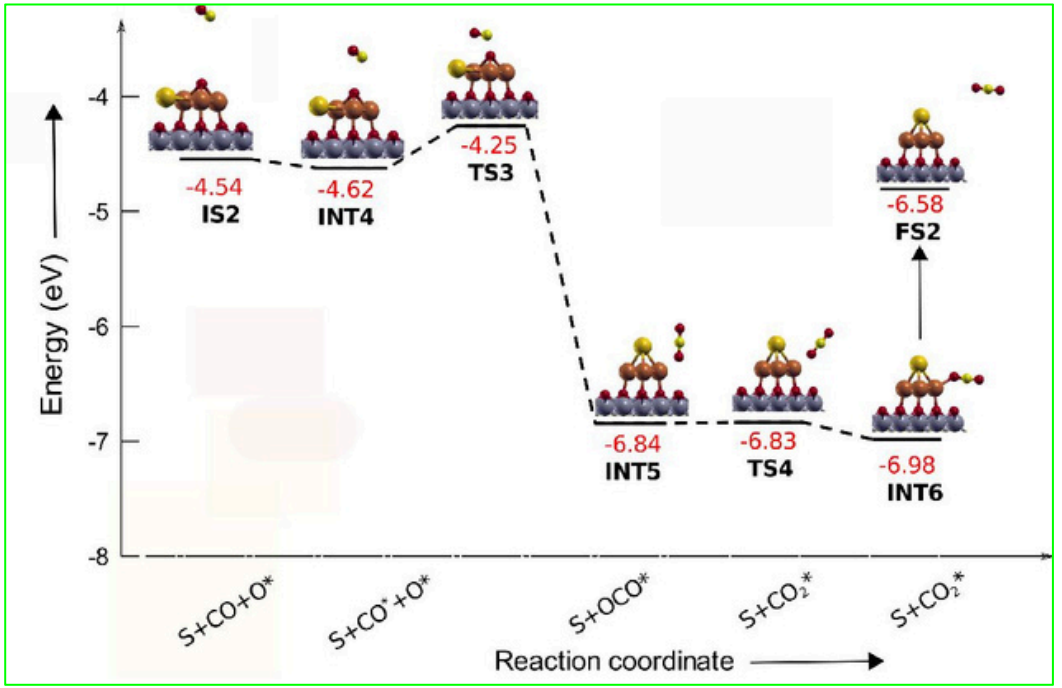
3) MvK – firstly CO adsorbs on the cluster and then it attacks an O atom from the Ti_2CO_2 MXene.



Thermally activated process: Example



FIRST CYCLE (LH)

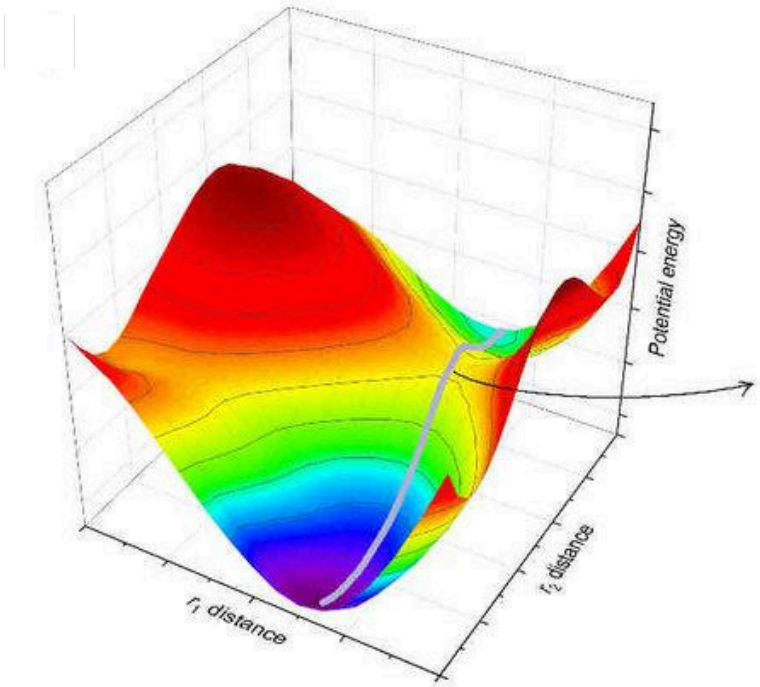


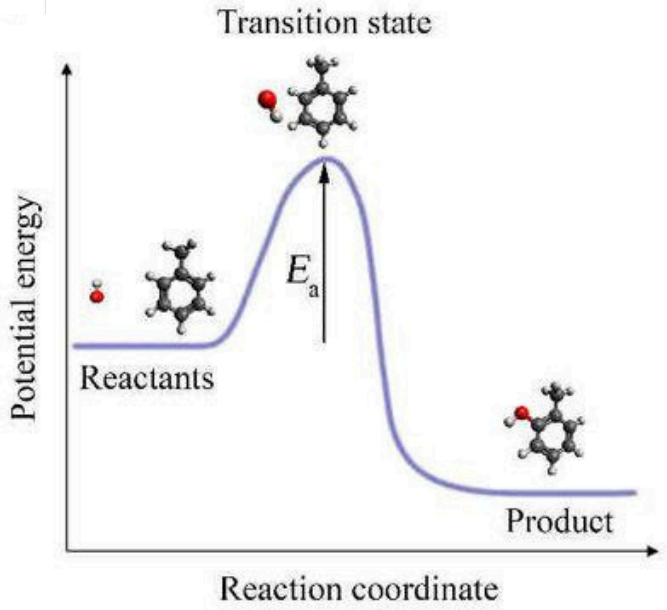
SECOND CYCLE (ER)

Potential energy surface

Reaction energy, $\triangle E = E_{FS} - E_{IS}$

Activation energy, $E_a = E_{TS} - E_{IS}$





Ecological Chemistry and Engineering S Volume 28, Issue 1 (March 2021)

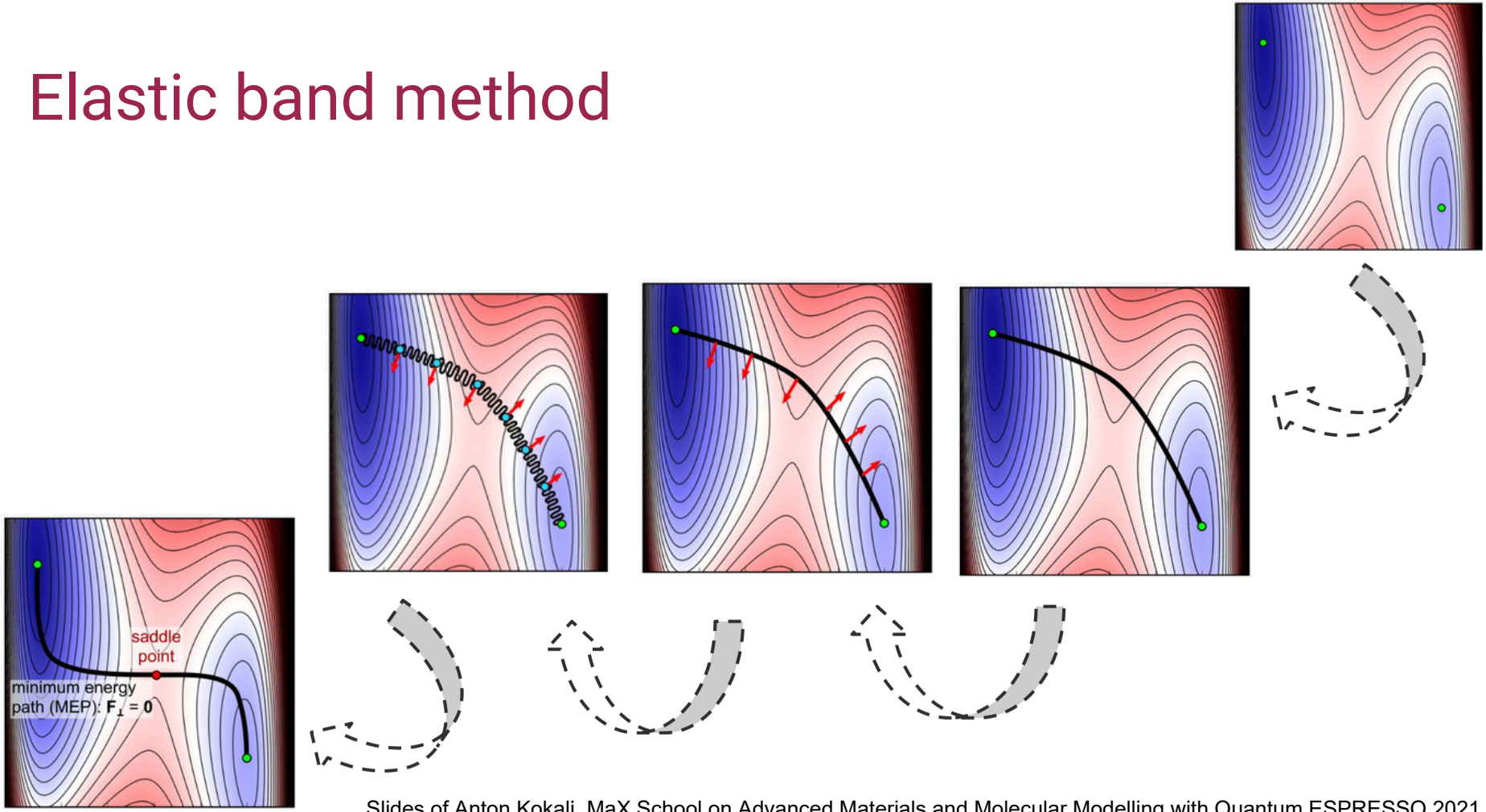
Activation energy

$$k = \frac{\prod_{j=1}^{3N} \omega_j^{in}}{\prod_{j=1}^{3N-1} \omega_j^{*}} e^{-(E^{*}-E^{in})/k_B T}$$

- Activation energy directly correlates to the rate constant

Activation barrier E_a (eV)	k (s^{-1}) — with $A = 10^{13}$
0.4 eV	$\sim 1.7 \times 10^{6}$ s^{-1} (instant)
0.6 eV	$\sim 7.2 \times 10^{2}$ s^{-1} (sub-second)
0.8 eV	$\sim 3.0 \times 10^{-1}$ s^{-1} (few seconds)
1.0 eV	$\sim 1.3 \times 10^{-4}$ s^{-1} (hours)
1.2 eV	$\sim 5 \times 10^{-8}$ s^{-1} (years)

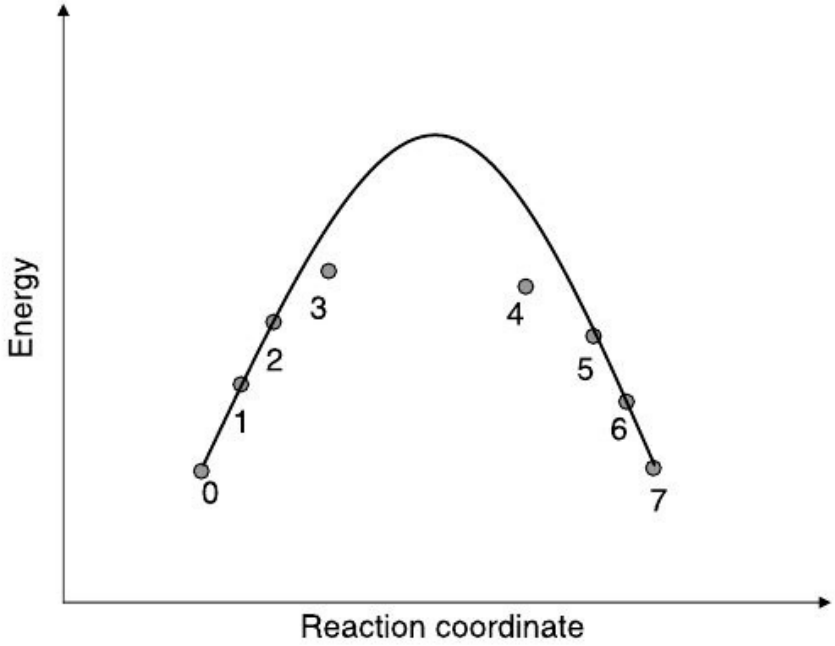
Elastic band method

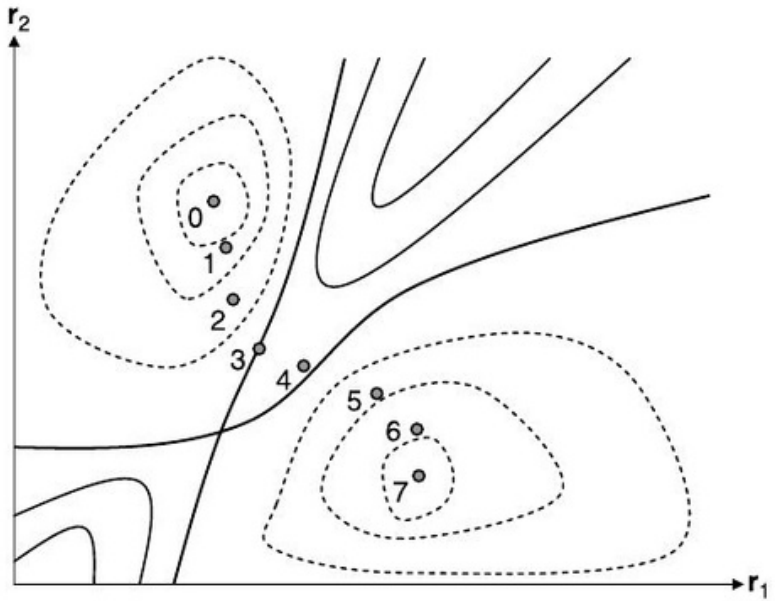


Slides of Anton Kokalj, MaX School on Advanced Materials and Molecular Modelling with Quantum ESPRESSO 2021

Problems: Sliding down

Corner cutting





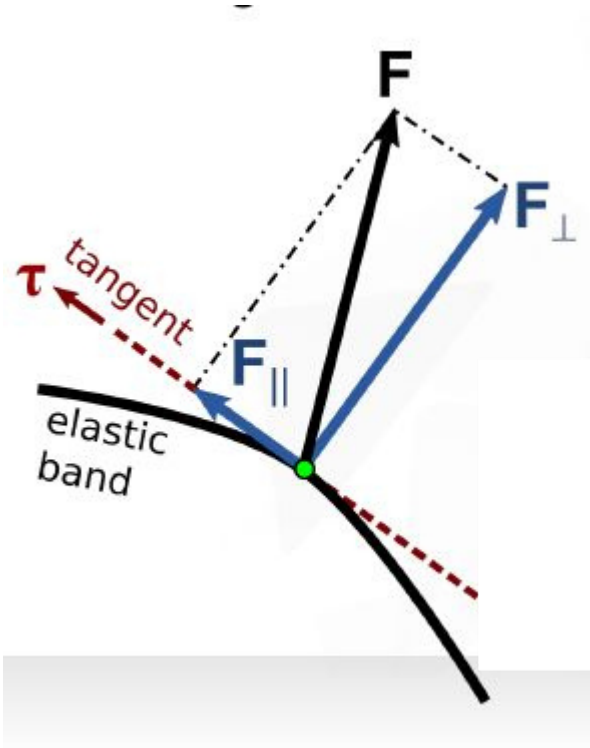
Density Functional Theory: A Practical Introduction, David S. Sholl, Janice A. Steckel

Solution: Nudging

$$\boldsymbol{F}(\boldsymbol{R}_i) = \boldsymbol{F}^s_{\parallel}(\boldsymbol{R}_i) + \boldsymbol{F}_{\perp}(\boldsymbol{R}_i), \; where$$

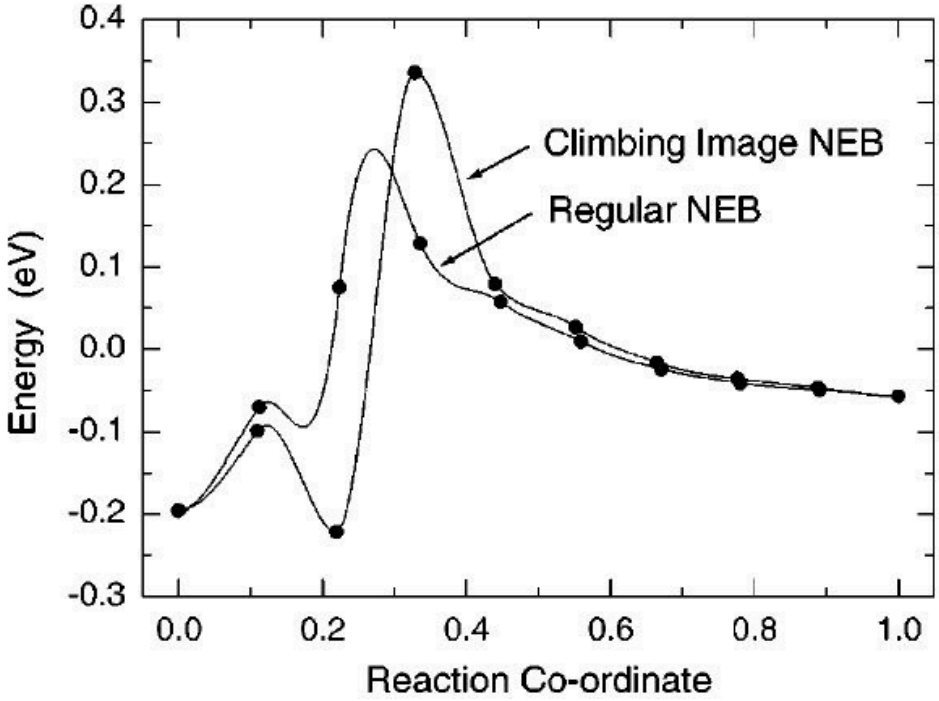
$$\boldsymbol{F}_{\perp}(\boldsymbol{R}_i) = -\nabla E(\boldsymbol{R}_i) + \nabla E(\boldsymbol{R}_i).\hat{\boldsymbol{\tau}}_{\boldsymbol{i}}\hat{\boldsymbol{\tau}}_{\boldsymbol{i}} \; and$$

$$\boldsymbol{F}^s_{\parallel}(\boldsymbol{R}_i) = k(2\boldsymbol{R}_i - \boldsymbol{R}_{i+1} - \boldsymbol{R}_{i-1})\hat{\boldsymbol{\tau}}_{\boldsymbol{i}}.$$



Density Functional Theory: A Practical Introduction, David S. Sholl, Janice A. Steckel

Climbing image

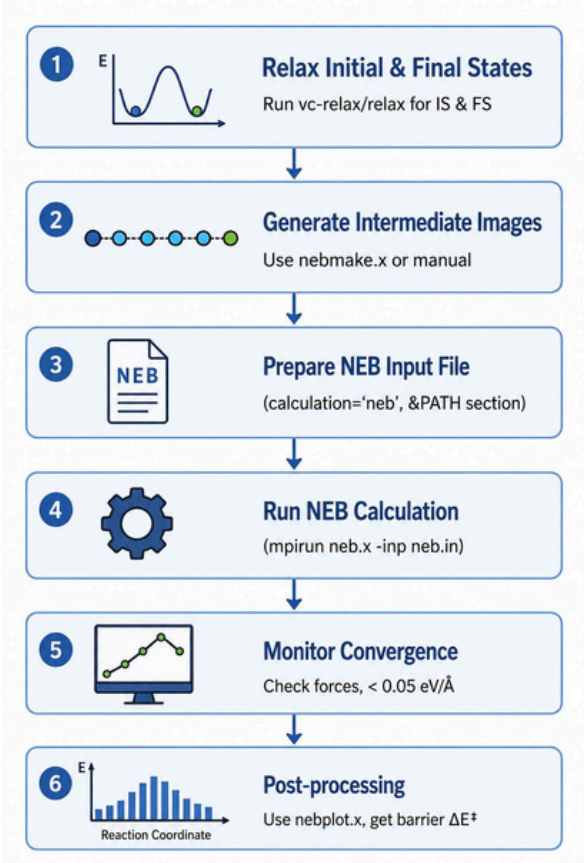


- Helps with the issue of transition state underestimation.

$$\boldsymbol{F}(\boldsymbol{R}_{im}) = -\nabla E(\boldsymbol{R}_{im}) + 2\nabla E(\boldsymbol{R}_{im}).\hat{\boldsymbol{\tau}_{im}}\hat{\boldsymbol{\tau}_{im}}$$

J. Chem. Phys. 113, 9901 (2000);

NEB workflow



Take-home advice

- Use intermediate images based on chemical intuition
- Decide number of images based on Inter-image distance (1-1.5 bohr is good enough)
- Start climbing image only when close to convergence
- Keep the order of atoms same in the coordinates for all images

Thank you